Datasheet DHT11

Temperature and humidity sensor

- Excellent impact resistance and electrical properties
- Full calibration
- Digital output, One-wire communication
- Excellent long-term stability
- Fast response, fast recovery time and strong anti-interference ability

Summary

DHT11 digital temperature and humidity sensor is a temperature and humidity composite sensor with calibrated digital signal output. It adopts the integrated digital temperature and humidity chip independently developed by Aosong electronic. It applies special digital module acquisition technology and temperature and humidity sensing technology to ensure good reliability and long-term stability of the product. The sensor is small, including a temperature and humidity composite sensor and a high performance 8-bit microcontroller, good consistency, high precision, rapid response, strong anti-interference ability.

Application

DHT11 is widely used in HVAC, dehumidifier, agriculture, cold chain storage, testing equipment, consumer goods, automotive, automatic control, data logger, weather station, household appliances, humidity regulator, medical, and other related humidity detection and control.

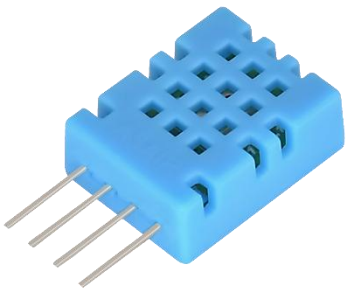


Figure 1. DHT11 Temperature and humidity sensor

1. T&RH sensor performance

1.1 Relative humidity

Table 1. Humidity characteristic table

No.	Parameter	Conditions	Min	Typical	Max	Unit
1	Scale range	Extended[1]	5	-	95	% RH
2	Accuracy[2]	-	-	±5		% RH
3	Repeatability	-	-	±1	-	% RH
4	Interchanegeability	-	Completely interchangeable			
5	Response time[3]	τ(63%)	-	<6	-	s
6	Hysteresis	-	-	±0.3	-	% RH
7	Drift typical value[4]	typical	-	<0.5	-	% RH/yr

1.2 Electrical Characteristics

Table 2. Electrical characteristics table

No.	Parameter	Conditions	Min	Typical	Max	Unit
1	The power supply voltage	-	3.3	5	5.5	V
2	Average current[5]	Sleep	-	60	-	μA
		Measure	-	1000	-	μA
3	Low level output voltage	I_L[6]=5mA	0	-	300	mV
4	High level output voltage	Rp<25kΩ	0.9*VCC	-	VCC	V
5	Low level input voltage	Falling down	0	-	0.3	V
6	High level input voltage	Rising up	0.7	-	VCC	V
7	R_{pu}[7]	VCC = 5V	1	4.7	100	kΩ
8	Sampling period	-	2	-	-	s

Electrical characteristics, such as power consumption, input and output high and low level voltages, depend on the power supply voltage. In order for sensors to communicate smoothly, it is important to ensure that the signal design is strictly limited to the range given in Table 2.

[1] Normal operating range: 8~85% RH, beyond this range, the sensor reading will be biased (drift > 3% RH after 60 hours at 90% RH). The working range is further limited to -20~60℃.

[2] This accuracy is the test accuracy of the sensor under the condition of 25℃ supply voltage of 5V during the factory inspection. This value excludes hysteresis and nonlinearity and applies only to non-condensing conditions.

[3] At 25℃ and 1m/s airflow, the time required to achieve the first-order response is 63%.

[4] This index is the result of clean air test at 25℃.

[5] Both the minimum and maximum average current are based on the condition that VCC = 5V and T = 25° C. The data of average current takes the average of 3 measurements.

[6] Low level output current.

[7] Represents the pull-up resistance, which value is generally 4.7kΩ.The longer the sensor lead is, the smaller the Rp should be. Please adjust the Rp according to your needs.

1.3 Temperature

Table 3. Temperature characteristic table

No.	Parameter	Conditions	Min	Typical	Max	Unit
1	The Precision of error	-	-	±2	-	℃
2	Repetitive	-	-	±1	-	℃
3	Hysteresis	-	-	±0.3	-	℃
4	Response time[8]	τ(63%)	5	-	30	s
5	Working range	-	-20	-	60	℃
6	Long time drift	-	-	< 0.5	-	℃/yr

[8] The response time depends on the rate of heat exchange between the air flow and the sensor.

2. DHT11 user guide

2.1 Working conditions

DHT11 used in the "Normal range" as shown in Figure 2 has the typical accuracy characteristics in Tables 1 and 3. Prolonged exposure to conditions outside the normal range, especially at humidity > 80%RH, may cause temporary signal drift (drift +3%RH after 60 hours). After returning to normal working conditions, the sensor will slowly self-recover to the correction state. See "Recovery Processing" in Section 2.2.2 to speed up the recovery process. Long-term use under abnormal conditions will accelerate the aging of the product.

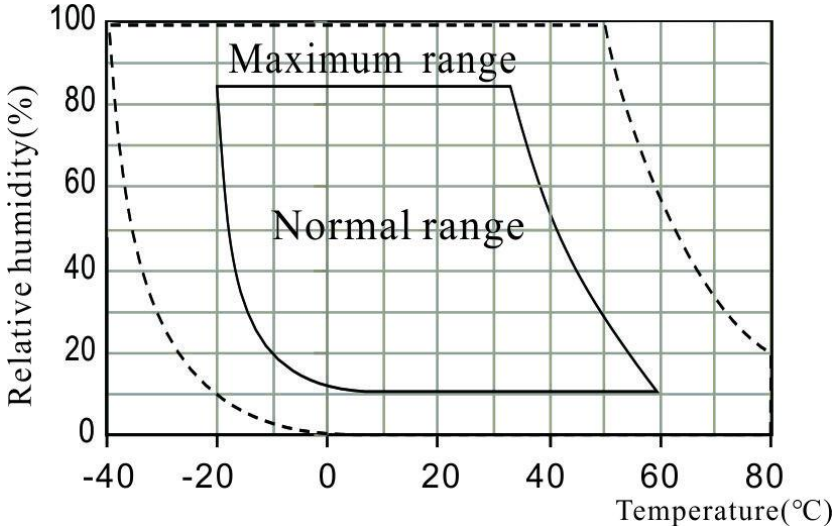


Figure 2. DHT11 Working conditions

2.2 Application information

2.2.1 Storage conditions and operation instructions

Humidity sensitivity level (HSL) is 1, according to IPC/JEDECJ-STD-020. Therefore, it is recommended to use within one year after shipment.

Temperature and humidity sensors are not ordinary electronic components, which need careful protection, which users must pay attention. Long-term exposure to high concentrations of chemical vapors can cause sensor readings drift. Therefore, it is recommended to store the sensor in the original package, including sealed ESD pocket, and meet the following conditions: temperature range of 10℃-50℃ ; Humidity is 20 to 60%RH (sensor without ESD package). For sensors that have been removed from their original packaging, we recommend storing them in anti-static bags made of metallic PET/AL/CPE materials.

Sensors should be protected from exposure to high concentrations of chemical solvents and prolonged exposure during production and transportation. Avoid contact with volatile glue, tape, stickers, or volatile packaging materials, such as foams. Production area should be well ventilated.

2.2.2 Recovery processing

As mentioned above, the readings can drift if the sensor is exposed to extreme working conditions or chemical vapors. It can be restored to the calibrated state by the following processing.

If the humidity is high, take drying measures: keep it at 60℃ and < 5%RH for 6 hours until it recovers.

If the humidity is low, the sensor can be re-hydrated. That is, the sensor can be stored at room temperature 25℃ and relative humidity > 75%RH for 24 hours to ensure the re-hydration of the polymer. Otherwise, the sensor reading will drift.

2.2.3 Temperature influence

The relative humidity of a gas is largely dependent on temperature. Therefore, when performing the test, ensure that the sensor under test is at the same temperature as the reference sensor, and then compare the

humidity readings.

2.2.4 Materials used for sealing and encapsulation

Many materials absorb moisture and act as buffers, which can increase response time and lag. Therefore, the material around the sensor should be carefully selected. Recommended materials are: metal material, LCP, POM (Delrin), PTFE (Teflon), PE, PEEK, PP, PB, PPS, PSU, PVDF, PVF.

The material used for sealing and bonding(Conservative recommendation): It is recommended to use epoxy filled methods for packaging electronic components, or silicone. Gases released from these materials also have the potential to contaminate sensors (see 2.2.1). Therefore, the sensor should be finally assembled and placed in a well-ventilated place or dried at >50 ° C for 24 hours to allow it to release contaminating gases before encapsulation.

3. Dimension

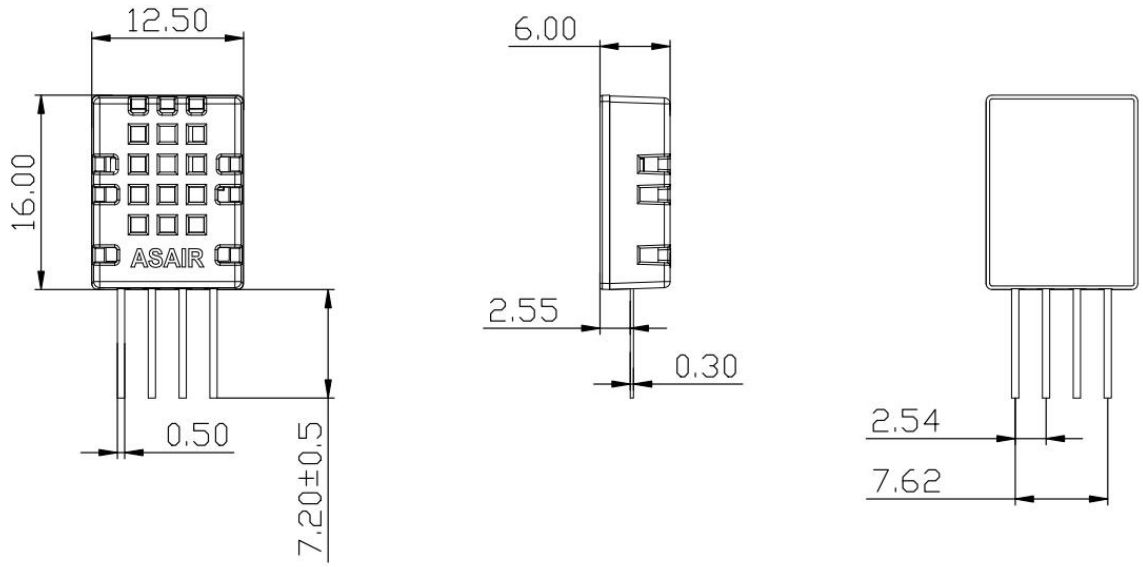


Figure 3. DHT11 Product size(Error ± 0.3mm)

4. Interface definition

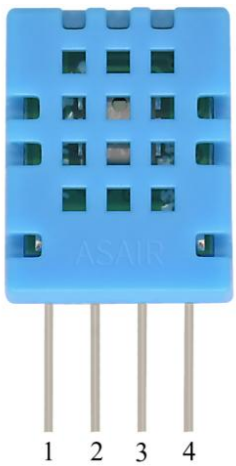


Figure 4. DHT11 Pin

Table 4. DHT11 pin Description

No.	Name	Description
1	VCC	Power supply 3.3 to 5.5V DC
2	SDA	Serial data, single bus
3	NC	Empty pin
4	GND	Ground, negative terminal of power supply

The electrical characteristics of DHT11 are defined in Table 2. Absolute maximum ratings are given in Table 5. Under such conditions, the functional operation of the device is not advisable. Prolonged exposure to the absolute maximum rating may affect the reliability of the sensor.

Table 5. DHT11 electrical Absolute maximum rating

Parameter	Min	Max	Unit
VCC to GND	-0.3	5.5	V
Digital I/O Pin (SDA) to GND	-0.3	VCC+0.3	V
Input current for each pin	-10	10	mA

ESD discharge complies with JEDEC JESD22-A114 standard (± 4kV in human mode) and JEDEC JESD22-A115 standard (± 200V in machine mode).

5. Sensor commumication

The DHT11 T/H sensor uses simplified single-bus communication. Single bus means only one data line, data exchange and control in the system are completed by the data line. The device (microprocessor) is connected to the data line through a drain open or tri-state port to allow the device to release the bus when it is not transmitting data, allowing other devices to use the bus; A single bus usually requires an external pull-up resistor of about 4.7kΩ, so that when the bus is idle, its state is high. Since they are master-slave structures, sensors respond only when the host calls them, so the host access to sensors must strictly follow the single-bus sequence, if there is sequence confusion, sensors will not respond to the host.

5.1 Typical One-wire circuit

The typical application circuit of the connection between the microprocessor and DHT11 is shown in Figure 5. The single-bus communication mode is composed of SDA pin of DHT11, pull-up resistor Rp and I/O port P3.0 of microprocessor (R5 and C4 are anti-jamming enhanced devices, R5=10Ω, Rp=4.7kΩ is recommended).

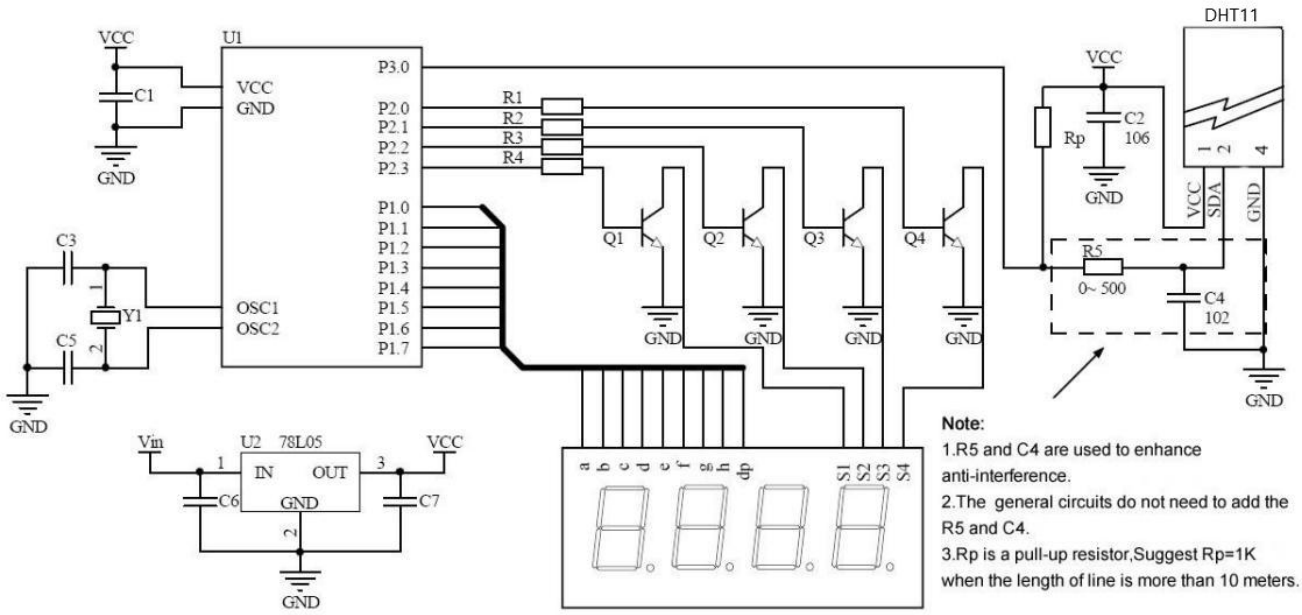


Figure 5. DHT11 One-wire Application Circuitry

Special instructions for single bus communication:

1. When 3.3V power supply is used, the length of connecting wire shall not be greater than 100cm. Otherwise, the line voltage drop will lead to insufficient power supply for the sensor, resulting in measurement deviation.

2. The minimum sensor reading interval is 2s. If the read interval is less than 2 seconds, the temperature and humidity may be incorrect, or the communication may fail. When the sensor is initially powered on, wait more than 2s before reading the sensor to get over the unstable state of the sensor.

3. The temperature and humidity value read out each time is the result of the last measurement. To obtain real-time data, read the temperature and humidity value twice consecutively. It is recommended that you read the sensor for multiple times consecutively with an interval of more than 2 seconds to obtain accurate data.

5.2 One-Wire bus transmission definition

SDA is used for communication and synchronization between micro controller and DHT11. It adopts single bus data format and transmits 40 bits of data at a time. The specific communication timing is shown in Figure.6, and the communication format is described in Table 6.

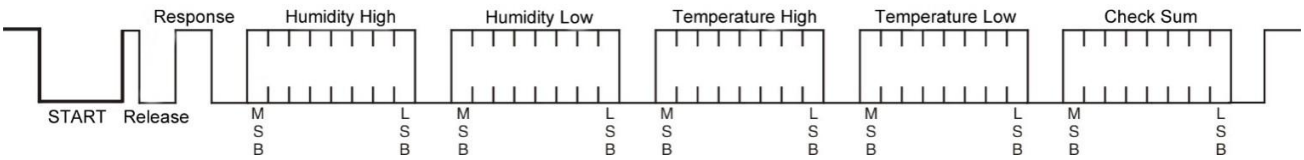


Figure 6. DHT11 One-wire bus transmission data graph

Table 6. DHT11 One-wire bus transfer data table

Name	Single bus format definition
START	The microprocessor pulls the data bus (SDA) down for a period of at least 18ms (maximum 30ms) to notify the sensor to prepare the data.
Response	The sensor pulls the data bus (SDA) low by 83μs and pills up by 87μs in response to the host's start signal.
Humidity	The high humidity indicates the integer part of the humidity(occupies 1 byte), and the low humidity indicates the decimal part of the humidity(occupies 1 byte).
Temperature	The high temperature is the integer part of the temperature data(occupies 1 byte), the low temperature is the decimal part of the temperature data(occupies 1 byte), and the low temperature Bit7 is 1, indicating the negative temperature, otherwise, the positive temperature.
Check sum	Check Sum = Humidity high bit+Humidity low bit+Temperature high bit+Temperature low bit

Example of data calculation:

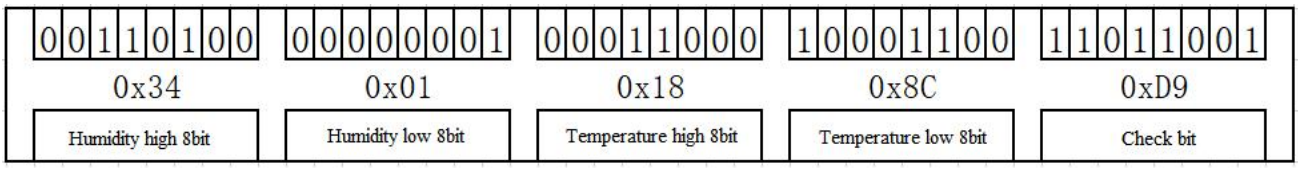


Figure 7. Examples for Single bus data calculation

When the data is transmitted as shown in Figure 7, check codes can be calculated and converted to obtain humidity and temperature according to the information in Table 6.

According to the calculation method of parity data in Table 6, the parity code can be obtained as follows:

34H+01H+18H+8CH=D9H

The calculated check code D9H is compared with the received check code. If they are the same, it means that the received humidity and temperature information data are correct; otherwise, the communication data should be discarded.

The values of humidity and temperature can be obtained from the conversion of data structures. For example, the high 8 digits (integer) of humidity is 34H, and the low 8 digits (decimal) of humidity is 01H. After converting the two digits into decimal, we can get 52.1, that is, the humidity is 52.1%RH. Similarly, it can be concluded that the temperature in Figure 7 is -24.12℃. The temperature here is negative because the highest bit of the low 8 bits of the temperature data, Bit7, is 1; When the most important bit, Bit7, is 0, the value is positive.

5.3 One-wire Communication Timing

Detailed timing signal characteristics are shown in Table 7, and the timing diagram of single-bus communication is shown in Figure 7. The temperature and humidity data read by the host from DHT11 is always the previous measurement value. If the interval between two measurements is long, read the temperature and humidity data twice consecutively. The second value is the real-time temperature and humidity value. In order to ensure the accurate communication of sensors, users should strictly follow the parameters and timing sequence in Table 7 and Figure 8 for design when reading signals.

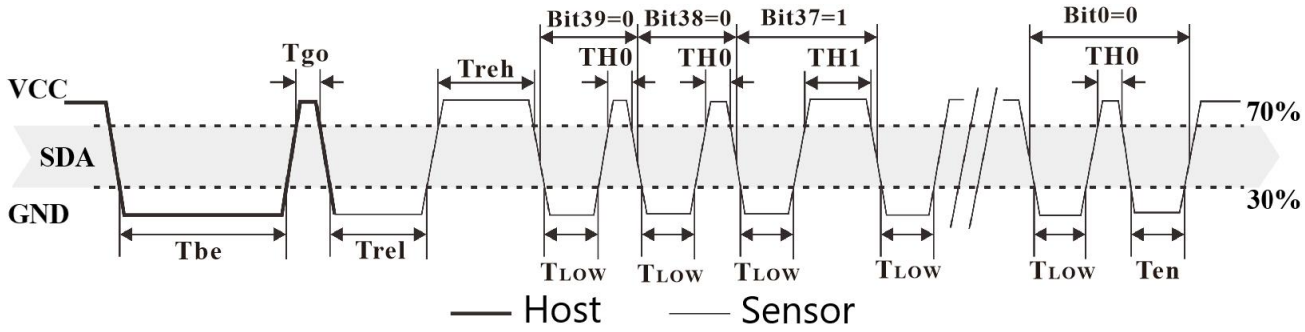


Figure 8. DHT11 Single Bus Communication Timing

Table 7. DHT11 timing signal characteristics

Symbol	Parameter	Min	Type	Max	Unit
T_{be}	Host start signal low time	18	20	30	ms
T_{go}	The host releases bus time	10	13	35	μs
T_{rel}	Response low time	78	83	88	μs
T_{reh}	Response high time	80	87	92	μs
T_{LOW}	Signal BitX= "0", "1" low level time, BitX: Bit0~Bit39	50	54	58	μs
T_{H0}	Signal BitX= "0" high level time	23	24	27	μs
T_{H1}	Signal BitX= "1" high level time	68	71	74	μs
T_{en}	The sensor releases bus time	52	54	56	μs

5.4 Peripheral Reading Process

When using a single bus to read the DHT11 sensor, the communication between the microprocessor and the DHT11 should complete the data reading process as shown in Figure 9. At the same time, our company also provides an example of reading code. Customers who need to download the code, please log in our website (www.aosong.com) to download the relevant code.

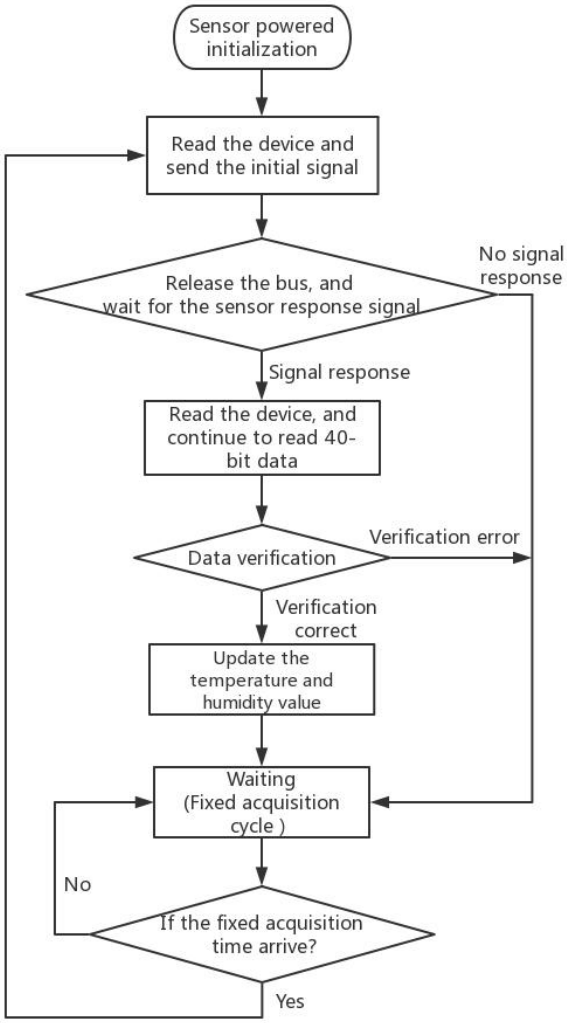


Figure 9. One-wire Bus read flow chart

After the DHT11 is powered on (wait for at least 2 seconds after the DHT11 is powered on to cross the unstable state, during which the reading device cannot send any command), test the ambient temperature and humidity data, and record the data. Then the sensor automatically goes to sleep. The SDA data line of DHT11 is pulled up by the pull-up resistor and keeps at a high level. At this time, the SDA pin of DHT11 is in the input state and detects external signals at all times.

I/O of microprocessor is set as output, and low level is output at the same time, and the low level holding time can not be less than 18ms (the maximum can not exceed 30ms), then I/O of microprocessor is set as input mode immediately after high level, release the bus, and after the host releases the bus, DHT11 sends a response signal. That is, the low output level of 80μs is used as the response signal, followed by the high output level of 87μs to notify the peripheral to prepare to receive data. The signal transmission is shown in Figure 10.

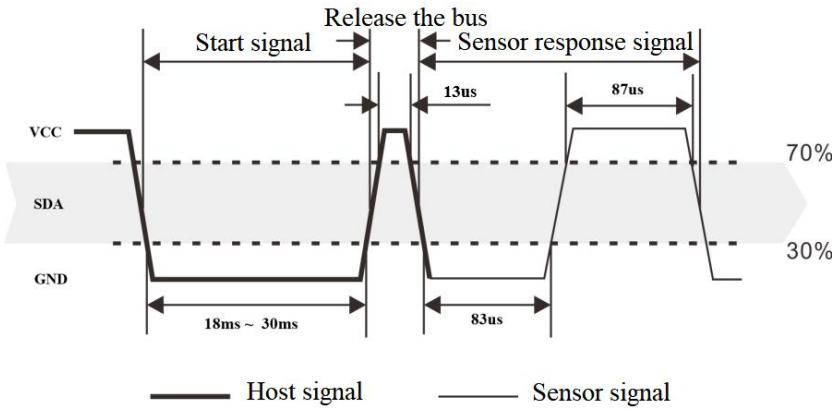


Figure 10. Response diagram

After DHT11 sends the response, the data bus SDA continuously outputs 40 bits of data, and the microprocessor receives 40 bits of data according to the change of I/O level.

The format of bit data "0" is: low level of 50-58μs and high level of 23-27μs;
The format of bit data "1" is: low level of 50-58μs and high level of 68-74μs;

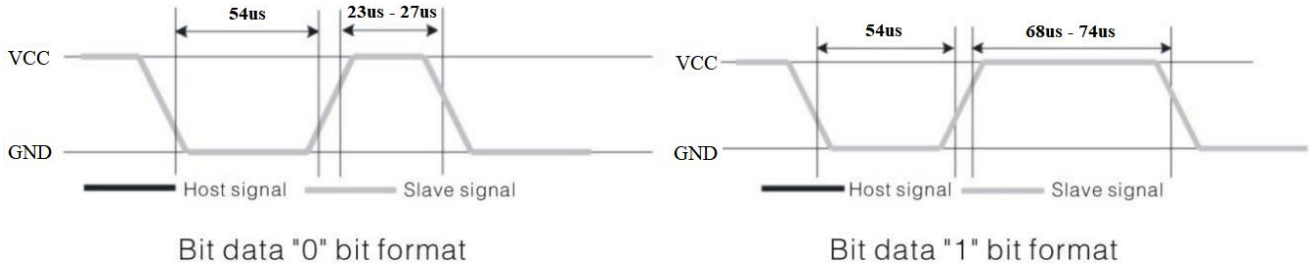


Figure 11. DHT11 Schematic diagram of sending data bits "0" and "1"

After the DHT11 outputs 40bits of data, it will continue to output 50 to 58μs of low level and then turn to the input state. The single bus is no longer controlled to release the single bus. This communication is complete.

After the communication is complete, the DHT11 collects temperature and humidity data once and automatically enters the sleep state. The DHT11 is awakened into single-bus communication mode only after receiving the initial signal from the microprocessor again. At the same time, the microprocessor will verify the received data according to the method in Section 5.2. If the verification is correct, the data will be analyzed to obtain the temperature and humidity value; otherwise, the data received in this communication will be discarded. After the end of the communication, the microprocessor can send the initial signal again to read the DHT11 after at least 2 seconds interval.

6. Stability of environment

If the sensor is used in equipment or machinery, make sure that the sensor used for measurement senses the same conditions of temperature and humidity as the sensor used for reference. If the sensor is placed in the equipment, the reaction time will be extended, so ensure that enough measurement time is allowed in the programming. The DHT11 sensor is tested according to the Aosong T&RH sensor enterprise standard. The performance of the sensor under other test conditions is not guaranteed and cannot be considered as part of the sensor performance. Especially to the user requirements of the specific occasion, do not make any commitment.

7. Packaging & Tracking information

All DHT11 sensors have laser markings on the back, see Figure 12. The carton is also labeled, as shown in Figure 13, and provides additional tracking information.

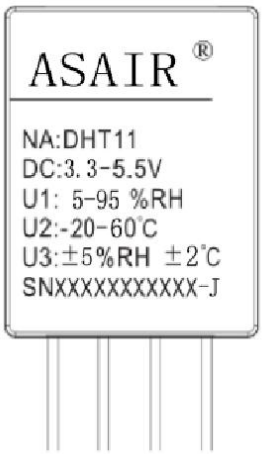


Figure 12. DHT11 sensor laser identifier

Warning and personal injury

Do not apply this product to safety protection devices or emergency stop equipment, and any other applications that may cause personal injury due to the product's failure. Do not use this product unless there is a special purpose or use authorization. Refer to the product data sheet and application guide before installing, handling, using or maintaining the product. Failure to follow this recommendation may result in death and serious personal injury.

The Company will not bear any compensation for personal injury and death arising therefrom, and will exempt the company's managers and employees, affiliated agents, distributors and any other claims that may arise therefrom, including: various costs, claims, lawyer fees, etc.

Quality assurance

Guangzhou Aosong Electronic Co., Ltd. provides the following quality guarantee to the direct buyers of its products (from the date of delivery), based on the technical specifications in the data manual of the products published by Aosong as the standard. Within the warranty period, if the product is confirmed to be defective, the company will provide free repair or replacement.

Description of warranty period

Accessories category	Shelf life
DHT11 Temperture and humidity sensor	12 months

The company is only responsible for products that are defective when used in applications that meet the technical conditions of the product. The company does not make any guarantees or written statements about the application of its products in those special applications. At the same time, the company does not make any promises about the reliability of its products when applied to products or circuits not provided by Aosong.

This manual may be changed at any time without notice.

Guangzhou Aosong Electronic Co., Ltd. reserves the right of final interpretation of this product.

Copyright ©2022, ASAIR ®